EXPERIMENT – 1

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AIM: Implementation of Reinforcement Learning System: Agent, Environment, State, Actions, and Reward.

1.

```
import random

class Environment:
    def __init__(self):
        self.state = 0

    def reset(self):
        self.state = 0
        return self.state

    def step(self, action):
        self.state += action
        if self.state == 5:
            reward = 10
            done = True
        else:
            reward = 0
            done = False
        return self.state, reward, done

class Agent:
    def __init__(self):
        self.actions = [-1, 1]

    def select_action(self):
        return random.choice(self.actions)

env = Environment()
agent = Agent()

state = env.reset()
done = False
step_count = 0

print("Starting episode.....")

while not done:
    action = agent.select_action()
```

```
    next_state, reward, done = env.step(action)
    print(f"Step {step_count}: State={state}, Action = {action}, Next State = {next_state}, Reward = {reward}")
    state = next_state
    step_count += 1
print("Episode finished")
```

OUTPUT:

```
PS C:\Users\aiml-lab\Desktop\RL> python exp1.py
Starting episode.........
Step 0: State=0, Action = -1, Next State = -1, Reward = 0
Step 1: State=-1, Action = 1, Next State = 0, Reward = 0
Step 2: State=0, Action = -1, Next State = -1, Reward = 0
Step 3: State=-1, Action = 1, Next State = 0, Reward = 0
Step 4: State=0, Action = 1, Next State = 1, Reward = 0
Step 5: State=1, Action = 1, Next State = 2, Reward = 0
Step 6: State=2, Action = 1, Next State = 3, Reward = 0
Step 7: State=3, Action = 1, Next State = 4, Reward = 0
Step 8: State=4, Action = 1, Next State = 5, Reward = 10
Episode finished
```

2.

```
import random

class Environment:
    def __init__(self):
        self.state = 0

    def reset(self):
        self.state = 0
        return self.state

    def step(self, action):
        self.state += action
        if self.state == 5:
            reward = 10
            done = True
        elif self.state == -5:
            reward = -5
            done = False
        else:
```

```
            reward = 0
            done = False
        return self.state, reward, done

class Agent:
    def __init__(self):
        self.actions = [-1, 1]

    def select_action(self):
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env = Environment()
agent = Agent()

state = env.reset()
done = False
step_count = 0

print("Starting episode.....")

while not done:
    action = agent.select_action()
    next_state, reward, done = env.step(action)
    print(f"Step {step_count}: State={state}, Action = {action}, Next State = {next_state}, Reward =
{reward}")
    state = next_state
    step_count += 1
print("Episode finished")
```

OUTPUT:

